Time series: linear forecasting

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The homework is provided on the final slides

Prediction equation in time domain

Suppose that we observe a realization of a **stationary** process with continuous or discrete time.

Let $x_{t_1}, x_{t_2}, \ldots, x_{t_n}$ be observed values, $t_1 < t_2 < \ldots < t_n$.

Consider the problem of predicting x_t, the value of the process at time t, $t \neq t_j$ for all j.

This problem is called

- **forecasting ahead**, if $t > t_n$;
- **backward forecasting**, if $t < t_1$;
- interpolation or **the reconstruction of missing values**, if $t_1 < t < t_n$.

The h-step ahead forecast

In practice, we usually have $t_1 = 1, \ldots, t_n = n$.

Let $x_1, x_2, \ldots, x_n$ be observed values of a weakly stationary process with mean μ and the covariance function $R(t)$.

The prediction of x_{n+h} is called h-**step ahead forecast**.

We consider the **linear estimator** of x_{n+h} in the form

$$\hat{x}_{n+h} = a_0 + a_1 x_1 + a_2 x_2 + \ldots + a_n x_n.$$

How to choose the coefficients $a_0, a_1, \ldots, a_n$?

MSE estimator

An estimator $\hat{x}_{n+h}$ is called the **mean square error** (MSE) **predictor** of x_{n+h} if it minimizes

$$\Psi(a_0, \ldots, a_n) = \mathrm{MSE}(\hat{x}_{n+h}) = \mathbf{E}(\hat{x}_{n+h} - x_{n+h})^2$$

$$= \mathbf{E}(a_0 + a_1 x_1 + a_2 x_2 + \ldots + a_n x_n - x_{n+h})^2$$

over all $a_0, a_1, \ldots, a_n \in \mathbf{R}$.

It is also called the "Best Linear Unbiased Predictor".

A solution of the problem of minimization of MSE can be found as a solution of the system of equations

$$\frac{\partial \Psi}{\partial a_0}(a_0, \ldots, a_n) = 0, \ldots, \frac{\partial \Psi}{\partial a_n}(a_0, \ldots, a_n) = 0.$$

Reformulation of MSE

$$\Psi(a_0,\ldots,a_n) =$$

$$= \mathbf{E}(a_0+a_1x_1+\ldots+a_nx_n-x_{n+h})(a_0+a_1x_1+\ldots+a_nx_n-x_{n+h})$$

$$= \mathbf{E}(a_0^2+2\sum_{j=1}^{n} a_0a_jx_j-2a_0x_{n+h}+\sum_{i=1}^{n}\sum_{j=1}^{n} a_ia_jx_ix_j-2\sum_{j=1}^{n} a_jx_jx_{n+h}+x_{n+h}^2)$$

$$= a_0^2+2\sum_{j=1}^{n} a_0a_j\mathbf{E}x_j-2a_0\mathbf{E}x_{n+h}+\sum_{i=1}^{n}\sum_{j=1}^{n} a_ia_j\mathbf{E}x_ix_j-2\sum_{j=1}^{n} a_j\mathbf{E}x_jx_{n+h}+\mathbf{E}x_{n+h}^2$$

$$= a_0^2+2\sum_{j=1}^{n} a_0a_j\mu-2a_0\mu+\sum_{i=1}^{n}\sum_{j=1}^{n} a_ia_j(R(j-i)+\mu^2)-2\sum_{j=1}^{n} a_j(R(n+h-j)+\mu^2)+R(0)+\mu^2$$

where $\mu = \mathbf{E}x_j$ is the mean of the stationary process and $R(j-i) = \mathrm{Cov}(x_i,x_j) = \mathbf{E}x_ix_j - \mu^2$ is the covariance function.

Solution of MSE criterion

A solution of the system of equations

$$\frac{\partial \Psi}{\partial a_0}(a_0, \ldots, a_n) = 0, \ldots, \frac{\partial \Psi}{\partial a_n}(a_0, \ldots, a_n) = 0.$$

is given by

$$a_0 = \mu\left(1 - \sum_{j=1}^{n} a_j\right) \quad \text{and} \quad a = (a_1, \ldots, a_n)^T = \Gamma^{-1} b_h$$

where $b_h = (R(h+n-1), \ldots, R(h+1), R(h))^T$ and

$$\Gamma = \Big(R(i-j)\Big)_{i,j=1}^{n} = \begin{pmatrix} R(0) & R(1) & R(2) & \ldots & R(n-1) \\ R(1) & R(0) & R(1) & \ldots & R(n-2) \\ R(2) & R(1) & R(0) & \ldots & R(n-3) \\ \vdots & \vdots & \vdots & & \vdots \\ R(n-1) & R(n-2) & R(n-3) & \ldots & R(0) \end{pmatrix}.$$

Variance of the MSE predictor

Th: The mean square error of the predictor $\hat{x}_{n+h}$ equals

$$\mathrm{MSE}(\hat{x}_{n+h}) = R(0) - b_h^T \Gamma^{-1} b_h.$$

Proof.

$$\mathrm{MSE}(\hat{x}_{n+h}) = \mathbf{E}(\hat{x}_{n+h} - x_{n+h})^2$$

$=$

$=$

where $X = (x_1, x_2, \ldots, x_n)^T$.

Prediction bounds

Suppose that $x_1, \ldots, x_n$ be a time series from a **stationary Gaussian (normal)** process. This means that $x_i \sim N(\mu, \sigma^2)$.

Therefore, we have $\hat{x}_{n+h} - x_{n+h} \sim N(0, \hat{\sigma}^2_{n+h})$,
where $\hat{\sigma}^2_{n+h} = \mathrm{MSE}(\hat{x}_{n+h})$.

Th: The **prediction bound** for the value of the process at time $n + h$ has the form

$$(\hat{x}_{n+h} - z_{1-\alpha/2}\hat{\sigma}_{n+h}, \hat{x}_{n+h} + z_{1-\alpha/2}\hat{\sigma}_{n+h})$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$-quintile of the standard normal distribution.

Note that $z_{1-0.05/2} = 1.96$.
Ⓡ *qnorm(1-0.05/2);*

Algorithm of computing the MSE predictor

Let $x_1, \ldots, x_n$ be a given time series from a stationary process with unknown mean and the given covariance function $R(k)$. Suppose that we want to construct h-step ahead forecast.

1. Compute $\hat{\mu} = (x_1 + x_2 + \ldots + x_n)/n$.
2. Construct the time series $y_i = x_i - \hat{\mu}$, $i = 1, \ldots, n$.
3. Compute the matrix Γ and the vector b_h.
4. Compute the vector $a = \Gamma^{-1} b_h$.
5. Compute $\hat{y}_{n+h} = a^T Y$, where $Y = (y_1, y_2, \ldots, y_n)^T$.
6. Compute $\hat{x}_{n+h} = \hat{\mu} + \hat{y}_{n+h}$ and $\hat{\sigma}^2_{n+h} = R(0) - b_h^T \Gamma^{-1} b_h$.

Note that y_i is obtained from x_i by shifting all values, so that the mean of the time series $y_1, \ldots, y_n$ is zero.

Algorithm of computing the MSE predictor, 2

Let $x_1, \ldots, x_n$ be a given time series from a stationary process with known mean μ and the given covariance function $R(k)$. Suppose that we want to construct h-step ahead forecast.

1. We do not estimate μ since μ is given.
2. Construct the time series $y_i = x_i - \mu$, $i = 1, \ldots, n$.
3. Compute the matrix Γ and the vector b_h.
4. Compute the vector $a = \Gamma^{-1} b_h$.
5. Compute $\hat{y}_{n+h} = a^T Y$, where $Y = (y_1, y_2, \ldots, y_n)^T$.
6. Compute $\hat{x}_{n+h} = \mu + \hat{y}_{n+h}$ and $\hat{\sigma}^2_{n+h} = R(0) - b_h^T \Gamma^{-1} b_h$.

Note that y_i is obtained from x_i by shifting all values, so that the mean of the time series y_i is zero.

Prediction of x_3 with $n = 2$

Task. Compute the predictor $\hat{x}_3$ for the time series $x_1 = 6$, $x_2 = 4$ and $R(k) = e^{-|k|}$.

Since we want to compute $\hat{x}_3$ and the last observed time moment is $n = 2$, we have $h = 1$. Then we obtain $\hat{\mu} = 5$, $y_1 = 1$, $y_2 = -1$.

$$\Gamma = \begin{pmatrix} R(0) & R(1) \\ R(1) & R(0) \end{pmatrix} = \qquad , \quad b_1 = \begin{pmatrix} R(2) \\ R(1) \end{pmatrix} =$$

$$\Gamma^{-1} = \qquad , \quad a = \Gamma^{-1} b_1 =$$

$$\hat{y}_3 = a^T Y =$$

$$\hat{x}_3 = \hat{\mu} + \hat{y}_3 =$$

Prediction of x_4 with $n = 2$

Task. Compute the predictor $\hat{x}_4$ for the time series $x_1 = 6$, $x_2 = 4$ and $R(k) = e^{-|k|}$.

Since we want to compute $\hat{x}_4$ and the last observed time moment is $n = 2$, we have $h = 2$. Then we obtain $\hat{\mu} = 5$, $y_1 = 1$, $y_2 = -1$.

$$\Gamma = \begin{pmatrix} R(0) & R(1) \\ R(1) & R(0) \end{pmatrix} = \qquad , \quad b_2 = \begin{pmatrix} R(3) \\ R(2) \end{pmatrix} =$$

$$\Gamma^{-1} = \qquad , \quad a = \Gamma^{-1} b_2 =$$

$$\hat{y}_4 = a^T Y =$$

$$\hat{x}_4 = \hat{\mu} + \hat{y}_4 =$$

Prediction of x_4 with $n = 3$

Task. Compute the predictor $\hat{x}_4$ for the time series $x_1 = 17$, $x_2 = 15$, $x_3 = 16$ and $R(k) = 1/(1 + \sqrt{|k|})$.

Here we have $n = 3$ and $h = 1$. Direct calculus gives
$\hat{\mu} = (x_1 + x_2 + x_3)/3 = 16$, $y_1 = 1$, $y_2 = -1$, $y_3 = 0$,

$$\Gamma = \begin{pmatrix} R(0) & R(1) & R(2) \\ R(1) & R(0) & R(1) \\ R(2) & R(1) & R(0) \end{pmatrix} =$$

$$b_1 = \begin{pmatrix} R(3) \\ R(2) \\ R(1) \end{pmatrix} =$$

$$\Gamma \cong$$

Inverse of matrix 3x3

For a matrix $\mathbf{A} \equiv \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, the inverse is

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \begin{bmatrix} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & \begin{vmatrix} a_{13} & a_{12} \\ a_{33} & a_{32} \end{vmatrix} & \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \\ \begin{vmatrix} a_{23} & a_{21} \\ a_{33} & a_{31} \end{vmatrix} & \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & \begin{vmatrix} a_{13} & a_{11} \\ a_{23} & a_{21} \end{vmatrix} \\ \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} & \begin{vmatrix} a_{12} & a_{11} \\ a_{32} & a_{31} \end{vmatrix} & \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}.$$

`http://www.wikihow.com/Inverse-a-3X3-Matrix`

Steps: $\det(A) = |A|$, transpose A, create minor matrices from A^T, change columns in minors for odd $i + j$, $\det$ of minors.

Prediction of x_4 with $n = 3$

$$\Gamma^{-1} = \begin{pmatrix} 1.4 & -0.55 & -0.31 \\ -0.55 & 1.55 & -0.55 \\ -0.31 & -0.55 & 1.4 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 0.37 \\ 0.41 \\ 0.5 \end{pmatrix},$$

$$a = \Gamma^{-1} b_1 =$$

$$\hat{y}_4 = a^T Y =$$

$$\hat{x}_4 = \hat{\mu} + \hat{y}_4 =$$

R script of the MSE predictor, 1

Task. Compute 1,2,3-ahead predictions and bounds for the time series 1,5,2,7,0,3,2,4 from a stationary process with unknown mean and $R(k) = 2.3e^{-0.5|k|}$.

```
® x=c(1,5,2,7,0,3,2,4); xpred=rep(0,3); xsig=rep(0,3);
givenCov=function(k) { return(2.3*exp(-0.5*abs(k))); }
n=length(x);
mu=mean(x);
y=x-mu;
Gamma=matrix(0,n,n);
for(i in (1:n))
for(j in (1:n))
  Gamma[i,j]=givenCov(i-j);
plot((1:n),x,xlim=c(1,n+5),pch=19,main=paste(x,collapse=","))
```

see next slide...

R script of the MSE predictor, 2

The continuation of the R script.

```
® for(h in (1:3)) {
  bh=rep(0,n);
  for(i in (1:n))
     bh[i]=givenCov(n+h-i);
  a=solve(Gamma) %*% bh;
  ynh=y%*%a; xpred[h]=mu+ynh;
  xsig[h]=sqrt(givenCov(0)-bh %*% solve(Gamma) %*% bh);
  points(n+h,xpred[h],col="red",pch=19)
  points(c(n+h,n+h),xpred[h]+xsig[h]*
      qnorm(1-0.05/2)*c(-1,1),col="blue",pch=19)
}
```

Homework with AI

Apply AI tools to deepen your understanding of linear forecasting and MSE predictors.

- **Conceptual Understanding:** Ask an AI to explain the "Best Linear Unbiased Predictor" (BLUP) property and why it is considered optimal for stationary processes.
- **Mathematical Verification:** Provide the MSE function $\Psi(a_0, \ldots, a_n)$ from Slide 5 to an AI and ask it to derive the system of linear equations for the coefficients a_j. Compare the output with the results on Slide 6.
- **R-Code Adaptation:** Copy the R-script from the slides into an AI and ask it to modify the code to calculate and plot a 95% confidence interval for h-step forecasts using a specific covariance function, such as $R(k) = 0.5^{|k|}$.

Homework

Practice on computing the inverse of 2×2 and 3×3 matrices by hand, that is needed for the exam. To check correctness, compute the inverse in the R package.

Practice on computing the 1-step and 2-step ahead prediction for time series of length 2 or 3 and a given covariance function $R(t)$ by hand. To check correctness, compute the predictions in the R package. Please, take any numbers for time series.

Examples of $R(t)$, which can be used in the exam, are

$$R(t) = 1.5e^{-1.3|t|}, \quad R(t) = 1.8e^{-1.7t^2},$$

$$R(t) = \frac{3}{1+1.4t^2}, \quad R(t) = 1.1e^{-0.9|t|^{1.5}}.$$

- Solve exam-like exercises for L6.